

姓名：

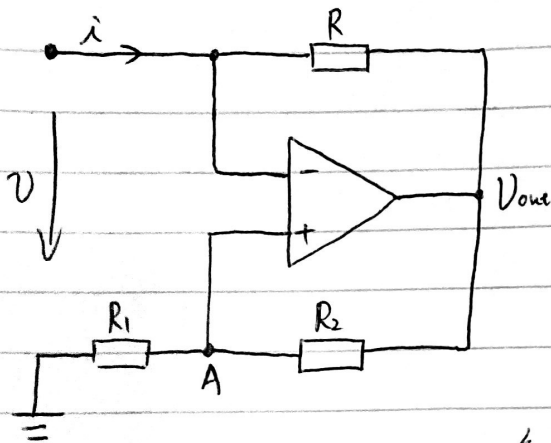
(號)

成績

年級：

科目：

作業1

當 $|V_{out}| < V_{sat}$ 時，OPA 工作在線性區

$$V_{out} = (1 + \frac{R_2}{R_1}) V$$

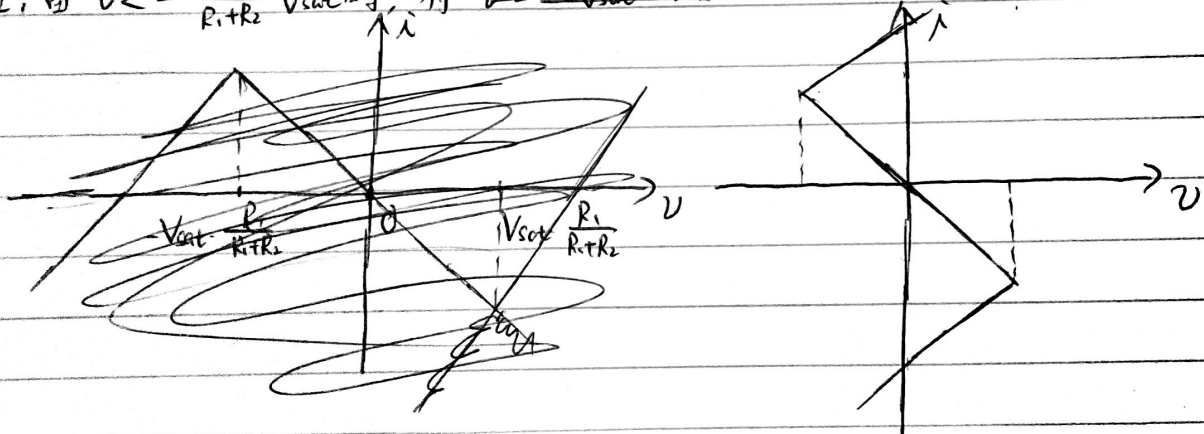
$$V - V_{out} = iR = -\frac{R_2}{R_1} V$$

$$\Rightarrow R_{in} = -R \frac{R_1}{R_2}$$

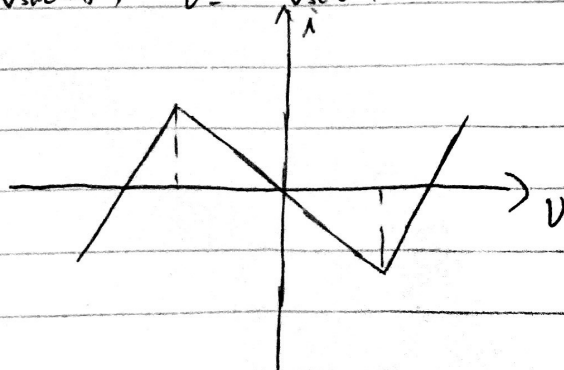
$$\text{此時 } |V| < \frac{R_1}{R_1 + R_2} V_{sat}$$

當 $V > \frac{R_1}{R_1 + R_2} V_{sat}$ 時，OPA 工作在負飽和區， $V_{out} = -V_{sat}$

$$\cancel{V = -V_{sat} + iR} \quad V = iR - V_{sat}$$

同理，當 $V < -\frac{R_1}{R_1 + R_2} V_{sat}$ 時，有 $\cancel{V = V_{sat} + iR} \quad V = V_{sat} + iR$ 

作業2

與上題相似，當 $|V| < \frac{R_1}{R_1 + R_2} V_{sat}$ 時，仍有 $R_{in} = -R \frac{R_1}{R_2}$ 當 $V > \frac{R_1}{R_1 + R_2} V_{sat}$ 時，OPA 工作在正飽和區， $V = V_{sat} + iR$ 當 $V < -\frac{R_1}{R_1 + R_2} V_{sat}$ 時， $V = -V_{sat} + iR$ 

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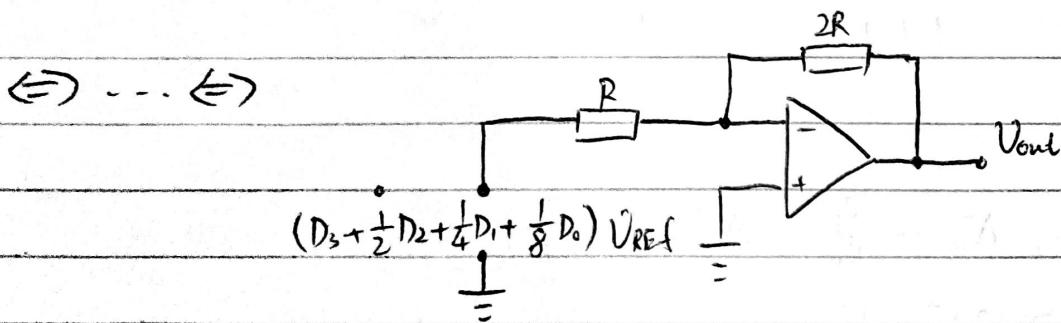
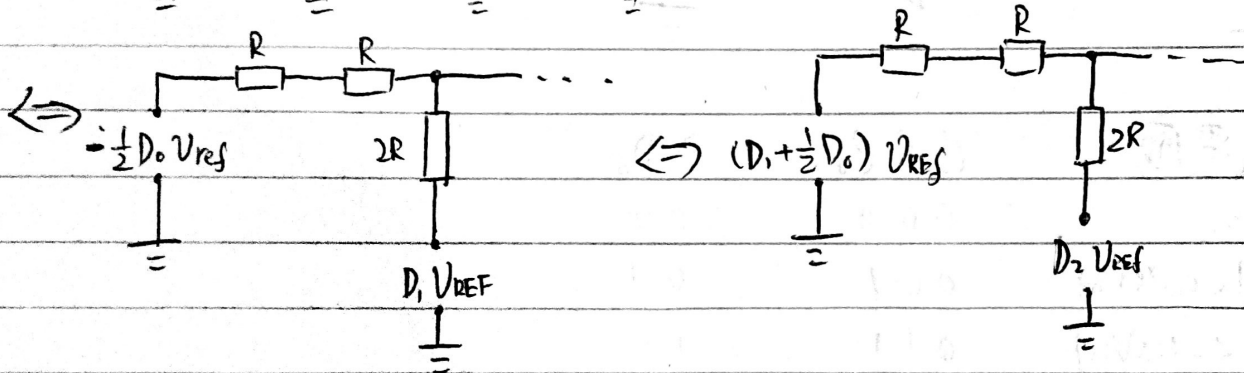
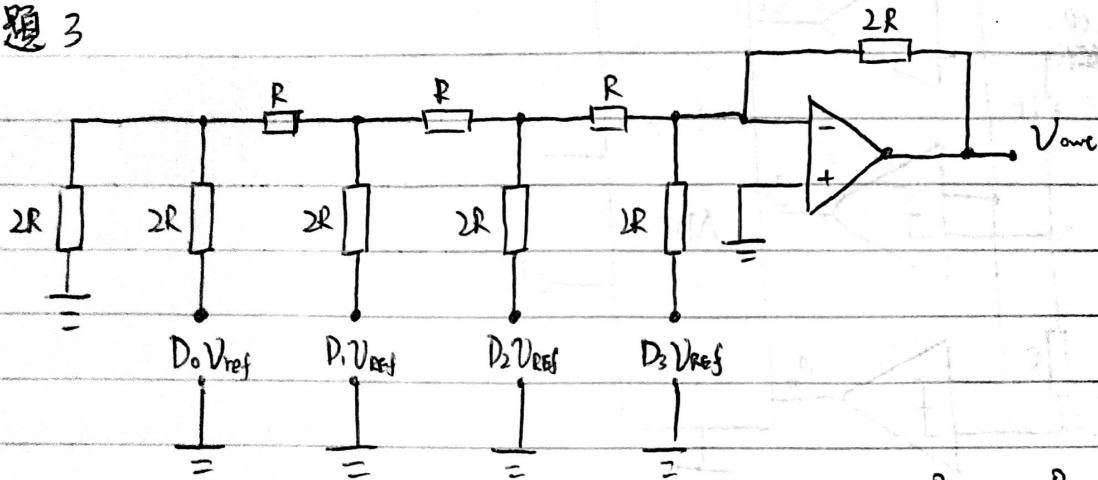
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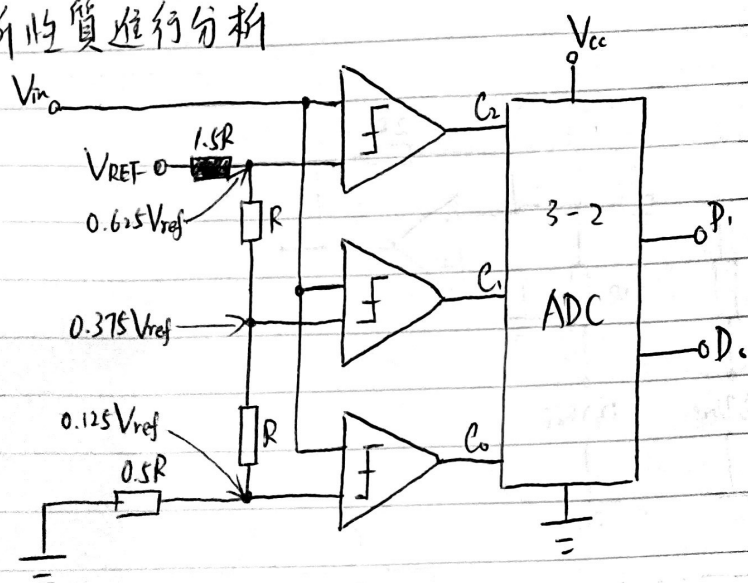
作業題 3



$$\frac{\left(\sum_{i=0}^3 \frac{1}{2^i} D_{3-i} \right) V_{REF}}{R} = - \frac{V_{out}}{2R} \Rightarrow V_{out} = - \left(\sum_{i=0}^3 \frac{1}{2^{i-1}} D_{3-i} \right) V_{REF}$$

作業4

利用虛斷性質進行分析



模擬輸入電壓	C_2, C_1, C_0	D_1, D_0
$< 0.125 V_{ref}$	0 0 0	0 0
$> 0.125 V_{ref}$ 且 $< 0.375 V_{ref}$	0 0 1	0 1
$> 0.375 V_{ref}$ 且 $< 0.625 V_{ref}$	0 1 1	1 0
$> 0.625 V_{ref}$	1 1 1	1 1

作業5

$$V_{out} = A_v^- V_{in}^- + A_v^+ V_{in}^+, \quad A_v^- = -\frac{R_2}{R_1}, \quad A_v^+ = \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{R_4}{R_3 + R_4}$$

$$\begin{cases} V_{id} = V_{in}^+ - V_{in}^- \\ V_{ic} = 0.5 (V_{in}^+ + V_{in}^-) \end{cases} \Rightarrow \begin{cases} V_{in}^+ = V_{ic} + 0.5 V_{id} \\ V_{in}^- = V_{ic} - 0.5 V_{id} \end{cases} \Rightarrow V_{out} = -\frac{R_2}{R_1} (V_{ic} - 0.5 V_{id}) + \left(1 + \frac{R_2}{R_1}\right) \frac{R_4}{R_3 + R_4} (V_{ic} + 0.5 V_{id})$$

$$\Rightarrow V_{out} = 0.5 \left(\frac{R_1 + R_2}{R_3 + R_4} \cdot \frac{R_4}{R_1} + \frac{R_2}{R_1} \right) V_{id} + \left(\frac{R_1 + R_2}{R_3 + R_4} \cdot \frac{R_4}{R_1} - \frac{R_2}{R_1} \right) V_{ic}$$

$$CMRR = \frac{1}{2} \left| \frac{\frac{R_1 + R_2}{R_3 + R_4} \cdot \frac{R_4}{R_1} + \frac{R_2}{R_1}}{\frac{R_1 + R_2}{R_3 + R_4} \cdot \frac{R_4}{R_1} - \frac{R_2}{R_1}} \right| = \frac{1}{2} \left| 1 + \frac{2 \cdot \frac{R_2}{R_1}}{\frac{R_1 + R_2}{R_3 + R_4} \cdot \frac{R_4}{R_1} - \frac{R_2}{R_1}} \right| = \frac{1}{2} \left| 1 + \frac{2 \cdot \frac{R_2}{R_1}}{\frac{R_1 + R_2}{R_3 + R_4} \cdot \frac{R_4}{R_1} - \frac{R_2}{R_1}} \right|$$

$$= \frac{1}{2} \left| 1 + \frac{2 \cdot \frac{R_2}{R_1}}{\frac{R_1 + R_2}{R_3 + R_4} \cdot \frac{R_4}{R_1} - \frac{R_2}{R_1}} \right| > \frac{1}{2} \left| 1 + \frac{2}{\frac{1 + \delta_4}{1 + \delta_2} - 1} \right| > \frac{1}{2} \left| 1 + \frac{2}{2\delta_{max} - 1} \right|$$

$$= \frac{1}{2} \left| \frac{2\delta_{max} - 1}{2\delta_{max}} \right|$$

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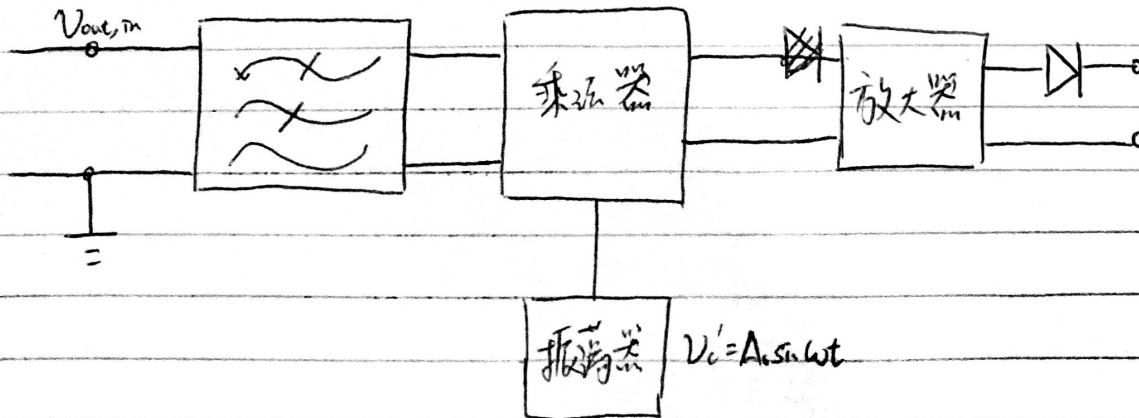
$$CMRR = \frac{1}{2} \left| \frac{(1 + \frac{R_2}{R_1}) \cdot \frac{R_4}{R_3+R_4} + \frac{R_2}{R_1}}{(1 + \frac{R_2}{R_1}) \cdot \frac{R_4}{R_3+R_4} - \frac{R_2}{R_1}} \right| = \frac{1}{2} \left| 1 + \frac{2 \frac{R_2}{R_1}}{(1 + \frac{R_2}{R_1}) \cdot \frac{R_4}{R_3+R_4} - \frac{R_2}{R_1}} \right|$$
$$= \frac{1}{2} \left| 1 + \frac{2}{(1 + \frac{R_1}{R_2}) \cdot \frac{R_4}{R_3+R_4} - 1} \right| = \frac{1}{2} \left| 1 + \frac{2}{(\underbrace{R_{10}, R_{20}}) \cdot (1+\delta_1)(1+\delta_2) \cdot \frac{1+\delta_4}{1+\delta_2} - 1} \right|$$
$$\geq \frac{1}{2} \left| 1 + \frac{2}{\left(\frac{1+\delta_{max}}{1-\delta_{max}}\right)^2 - 1} \right| = \frac{1}{2} \left| 1 + \frac{2}{\frac{2\delta_{max}}{1-\delta_{max}} \cdot \frac{2}{1-\delta_{max}}} \right| = \frac{1}{2} \left| 1 + \frac{(1-\delta_{max})^2}{2\delta_{max}} \right|$$
$$\approx \frac{1}{2} \left| 1 + \frac{1-2\delta_{max}}{2\delta_{max}} \right| = \frac{1}{4} \left| \frac{1}{\delta_{max}} \right| \quad (\delta_{max} \text{ 充分小}) = \frac{1}{4\delta_{max}}$$

" $CMRR_{min}$

$$\text{If } CMRR = CMRR_{\min} = 80\text{dB} \leq 20 \log \frac{1}{4\delta_{\max}} \Rightarrow \frac{1}{4\delta_{\max}} \geq 10^4$$

$$\Rightarrow \delta_{\max} \leq 0.25 \times 10^{-4} = 2.5 \times 10^{-5}$$

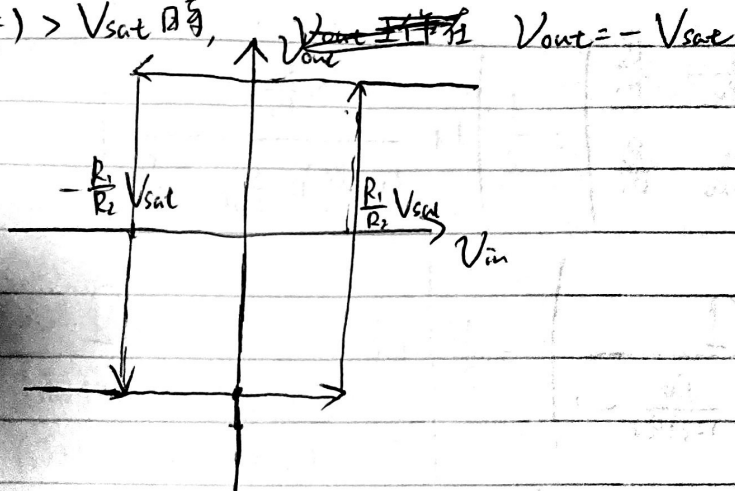
作業 6



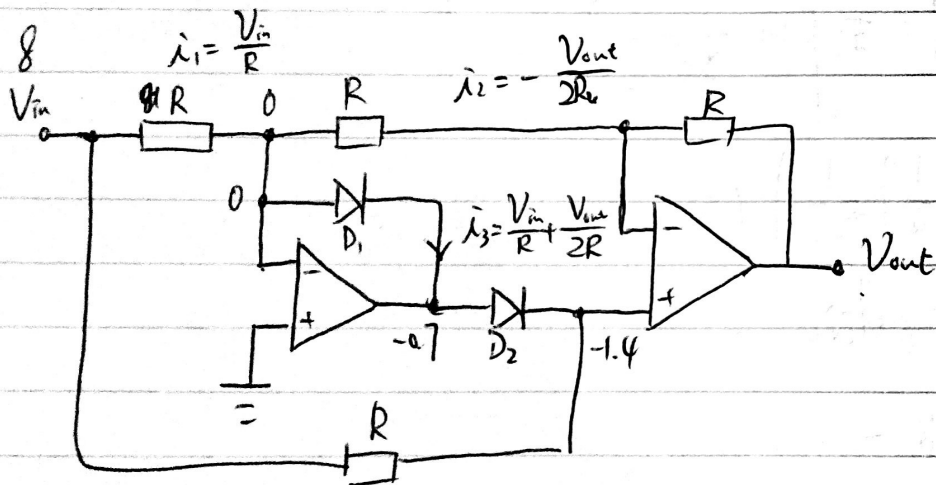
作業 7

同相 Schmitt 觸發器

當 $V_{in} \cdot (-\frac{R_1}{R_2}) > V_{sat}$ 時,



作業 8



當 $V_{in} > 0$ 時, D_2 截止,

$$\frac{0 - V_{out}}{2R} = \frac{0 - V_{in}}{R} \Rightarrow V_{out} = 2V_{in}$$

當 $V_{in} < 0$ 時, D_1, D_2 導通.

$$\frac{-1.4V}{R} = -\frac{V_{out}}{2R} \Rightarrow V_{out} = -2.8V$$

